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Structure of ecosystem:—

Composition and organization of biological communities and abiotic component constitute the structure of an ecosystem. The structure of ecosystem can be divided into two parts.

1. Biotic structure:—

The plants, animals and microorganism present in an ecosystem form biotic component. These organism have different nutritional behaviour and status in the ecosystem and are accordingly known as producers or consumers, based on how they get their food.

(a) Producers:—

They are mainly the green plants, which can synthesize their food themselves by making use of carbon dioxide present in the air and water in the presence of sunlight by involving chlorophyll through the process of photosynthesis.

Consumers:—

All organism which get their organic food by feeding upon other organism are called Consumers.

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2. Abiotic structure:-

The physical and chemical component of an ecosystem constitute its abiotic structure. It includes climatic factors, soil factors, geographical factors, energy, nutrient and toxic substances.

Physical factor:-

The sun light and shade, intensity of solar flux, duration of sun hours, average temperature, maximum minimum temperature, annual rainfall, wind, soil type, water availability, water current etc. are some of the important physical features which have a strong influence on the ecosystem.

Chemical factor:-

Availability of major essential nutrients like carbon, nitrogen, phosphorus, potassium, hydrogen, oxygen and sulphur, level of toxic substances etc.

Organic substances present in the soil or water largely influence the functioning of the ecosystem.

All the ~~bi~~ biotic components of an ecosystem are influenced by the abiotic component and vice versa.

Biotic:-

Biotic factors are living things within an ecosystem, such as plants, animals and bacteria etc.

abiotic:-

abiotic are non living components such as water, soil and atmosphere.

Aquatic ecosystem:-

Aquatic ecosystem include oceans, lakes, rivers, streams and wet lands. Within these aquatic ecosystem are living things that depend on the water for survival such as fish, plants and microorganism.

Lentic and lotic ecosystem:-

Lentic:-

Lentic ecosystem are still water ecosystem.

Eg:- Ponds, marshes, lakes etc.

Lotic:-

Lotic ecosystem are characterized by flowing water.

Example:- include rivers, streams and springs.

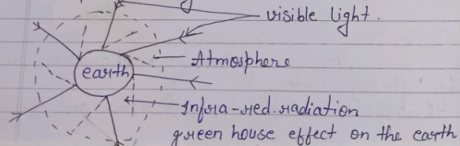
Terrestrial ecosystem:-

A terrestrial ecosystem is a land based community of organisms and the interactions of biotic and abiotic component in a given area. Example of terrestrial ecosystem include the tundra, taigas, temperate deciduous forests, tropical rainforests, grasslands and deserts.

Global warming or green house effect

The heating of the atmosphere due to absorption of infrared radiation by Carbon dioxide and other gases is called green house effect.

The gases which cause green house effect are Carbon dioxide, water vapour, Chlorofluoro Carbons and oxides of nitro-nitrogen. These gases in the atmosphere form a thick cover around the earth. Sun emits energy in the form of U.V radiations, visible light and I.R radiation. The ozone layer absorb most of the U.V radiation to pass through towards the earth.



Causes of green house effect

1. Burning of fossil fuel :-

Fossil fuels such as coal, oil and natural gases has now become an integral part of man's life. Upon burning, they release large amount of CO_2 in the atmosphere.

2. Deforestation

Green plants use CO_2 to prepare their food by photosynthesis and release O_2 . Thus helps in maintaining O_2 - CO_2 balance of the atmosphere. The large scale deforestation for urbanization and industrialization for urbanization and industrialization has resulted in the increase of concentration of CO_2 in atmosphere.

3. Increase in population

The population increase has also contributed to increase in the concentration of CO_2 .

4. Farming

We use fertilizer for farming. Nitrous oxide is one of the green house gas that is used in fertilizer.

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Industries

5. Cement industries are one of the major contributors of CO_2 which is most important green house gas. fertilizer industries, oil extraction and coal mining activities also contributed much to green house gases.

Wastes and Landfills

The garbage and other wastes produce carbon dioxide and methane gas upon decomposition.

Harmful effect of Green house Effect

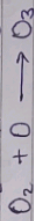
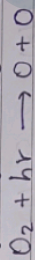
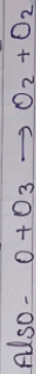
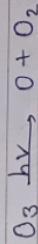
1. Due to high temperature glaciers and polar ice caps will melt resulting in flooding of coastal plain.
2. The fertile lands may become barren and aquatic life will be affected.
3. High temperature will reduce work efficiency of human beings.
4. High temperature is likely to increase the incidence of infectious diseases like malaria, dengue and yellow fever etc.

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5. Due to increased CO_2 in atmosphere, the rate of photosynthesis will increase and it may change rain fall pattern.

Ozone layer depletion

Ozone is a bluish gas. The stratosphere (upper parts of atmosphere) (11-50 km above earth's surface) contains ozone molecules which play a significant role in protecting earth from the U.V. radiation coming from the sun. Ozone absorbs U.V. radiation and prevent most of them in reaching the earth's surface.



Ozone is simultaneously being formed and destroyed by naturally occurring reaction.

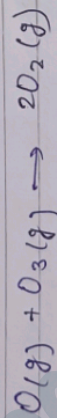
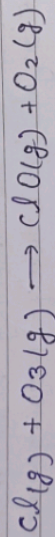
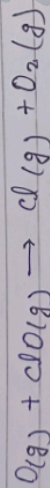
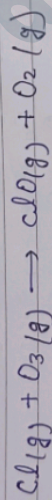
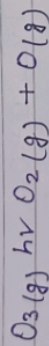
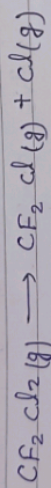
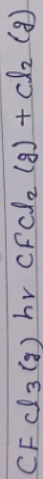
Cause of Depletion of ozone layer

Chlorofluoro Carbon (CFCs) are the exhausts by super sonic air craft and Jumbo Jet flying in the upper atmosphere. These got accumulated at high altitudes and upon go decomposition, under the influence of ultraviolet radiation, one of the main decomposition product is chlorine. Each atom

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of chlorine Cl_2 released with more than 10⁵ molecules of ozone converting ozone into oxygen. Consequently, gradual decomposition of ozone layer takes place and the affect capacity of ozone layer to stop ultraviolet radiation from entering the environment diminishes.

The typical reaction causing depletion of ozone layer by CFC to be:-



Effect of ozone layer depletion

With the depletion of atmospheric ozone layer the extent of U.V radiation over the earth's atmosphere increases. U.V radiation are harmful to man's life.

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They Cause Skin Cancer, swelling of skin, Sun burns, burning injury, disturbance of immune system of human body etc.

Unit - 3

Noise pollution

As unwanted, harsh and loud sound which reduces concentration and work efficiency is called noise. Noise produces illeffect on our health and hence is a pollutant like other pollutants, noise is also a by-product of industrialisation and modern civilisation.

The disturbance produced in our environment by the undesirable loud sound of various kinds is called noise pollution.

Sources of noise pollution

Some of the important sources of noise pollution in homes, industry and transport are given below.

(i) In homes:-

Some of the sources which produce noise pollution in homes are radio, television, mixer grinder, washing machine, vacuum cleaners, air conditioners etc. In addition to these loudspeakers and bands used during

procession, celebration of festival, marriage. They are sources of noise pollution.

(ii) In industry:-

Machines used in the industries make loud and irritating noises of different kinds which produce a lot of noise pollution.

(iii) In transport:-

The various means of transport create a lot of noise pollution. For example:- engines and horns of vehicles like scooter, motor-cycle, car, trucks, buses, trains produce a lot of noise in cities. The landing, take-off and flying of aeroplanes etc.

Measurement of noise

The unit of measurement of the intensity of sound (noise) in decibels, dB and is represented as -

$$1 \text{ decibel (dB)} = 10 \log \frac{\text{measured quantity of sound intensity}}{\text{Reference standard quantity of sound intensity}}$$

Human ear can tolerate noise up to 120 decibels. The range of audible sound to painful noise varies from 20 to 130 decibels.

Effect of noise pollution

1. Noise pollution reduces the hearing capacity of a person.
2. Noise causes head-ache, irritability and extreme emotional behaviour.
3. Noise increases nervous tension, blood pressure and heart trouble.
4. Noise pollution disturbs our peace of mind, sleep, studies as well as work.
5. Noise pollution causes impairment of night vision and decreases the state of colour-perception.
6. Noise interferes in communication with other person. For example, during rattling noise of a machine, we cannot hear properly even a person standing nearby.

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Control of noise pollution at sources

1. Restrict the use of loud-speakers and amplifiers to reduce noise pollution. television, and speakers should be played at low volume.
2. The industrial units should be setup away from the residential areas.
3. Machines should be lubricated from time to time and maintained properly.
4. Use better designed engines for transport vehicles (car, buses, trucks, train and aeroplanes) which make less noise.
5. Noise producing vehicles should be legally banned.

Control noise pollution by putting obstacles in the path of noise.

1. Trees should be planted along the roads and in the residential areas.
2. Acoustic material and mufflers should be used to absorb sound.

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Protect sound receptive organs (ears)

1. To reduce the harmful effect of loud noise, cotton plugs or ear muffs should be used.

Noise pollution rules 2000

To regulate and control noise producing and generating sources with the objective of maintaining the ambient air quality standards in respect of noise.

Air pollution

Air pollution is defined as the presence of excessive or unwanted foreign substances in the air which adversely affect the quantity of air and causing damage to human, animal and plants.

Natural sources of air pollution:-

- (i) volcanic eruptions release poisonous gases such as SO_2 , H_2S , CO etc.
- (ii) forest fires.
- (iii) Decay of vegetables matter.
- (iv) Dust and sandstorms.
- (v) Salt particles from oceans.
- (vi) Pollen grain of flowers.
- (vii) Bacteria, viruses etc.

Man-made sources of air pollution

- (i) High population density :- increase in population.
- (ii) Rapid industrialisation :- increase in industries.

(iii) Burning of fuels and fires :- Burning of petroleum product coal and coke.

(iv) Deforestation :- Cutting of trees.

(v) Emission from vehicles :- Vehicles emissions, containing dangerous gases such as CO , nitrogen oxide and hydrocarbons.

(vi) Agricultural activities :- Pesticides, insecticides etc. are used in agriculture.

(vii) Wars :- The nuclear weapons used during wars emit out radiations which adversely affect the health.

(viii) Combustion engines and power plants.

Air pollutionAir pollutants [Major]

- (i) CO and CO_2
- (ii) Oxides of sulphur
- (iii) oxides of nitrogen
- (iv) hydrocarbons.
- (v) Particulate like - metal particles, fly ash, dust etc.
- (vi) smoke.
- (vii) Smog - mixture of smoke and fog.

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Type of air pollutant(i) Primary air pollutant

These are the air pollutants that are directly released into the atmosphere from the sources. These include particulate matter such as ash, smoke, dust, fumes etc. inorganic gases such as sulphur dioxide, carbon monoxide, hydrogen sulphide, nitrogen oxide, ammonia, hydrogen fluoride and carbon dioxide, hydrocarbon and radioactive substance.

(ii) Secondary air pollutant :-

These are the air pollutant that are formed in the atmosphere by the chemical interactions (chemical reaction) among primary pollutant and the normal constituent of the atmosphere. Secondary air pollutant include sulphur trioxide, nitrogen dioxide, ozone, aldehyde, ketones, nitrate salts etc.

Harmful effect of air pollutant

1. The toxic effects of CO on human beings and animals arise from its reversible combination with haemoglobin of the blood due to which the oxygen

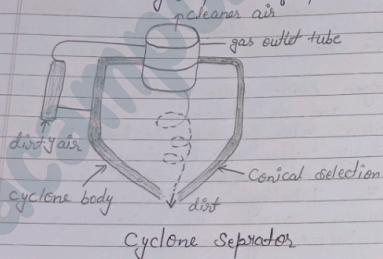
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carrying capacity of the blood decreases. The decreased availability of oxygen in body cells result in anoxia (oxygen starvation) and death may occur.

2. The high concentration of CO₂ in atmosphere is responsible for the rise of atmospheric temperature. (green house effect).
3. SO₂ irritates the respiratory system of animal and human. It also produces corrosive effect on building material such as lime stone, marble, roof slate etc.
4. Nitrogen oxide cause fading of textile dyes and deterioration of cotton and nylon fabric fabrics. Several plants are adversely affected by nitrogen oxide and start shedding their leaves and fruits. Inhaling of nitrogen oxide by humans result in pulmonary edema and haemorrhage.
5. Chlorofluorocarbons (CFC) which are widely used as propellants and refrigerants escape into the atmosphere. When they reach the upper sphere of the atmosphere they disturb the ozone layer.
6. The particulates seriously affect the human respiratory system and cause bronchitis, asthma, lung cancer and various allergic reactions.

Effect of particulate pollutant Control by cyclone separator

Cyclone separator are simply cyclones are separation devices (dry scrubbers) that use the principle of inertia to remove particulate matter from flue gases. Cyclone separator is one of many air pollution control devices known as pre-cleaners since they generally remove larger pieces of particulate matter.



Effect of particulate pollutant

Breathing in unhealthy level of particulate matter can increase the risk of health problem like heart disease, asthma and low birth weight. Unhealthy levels can also reduce visibility and cause the air to appear hazy.

Control of air pollution by electrostatic precipitator

An electrostatic precipitator collect fly ash, dust and soot particles from furnaces or boilers in thermal power plants, steel plants and various industrial plants. It is a type of device used in commercial industries to control air pollution.

The air (prevention and control of pollution) Act, 1981

The air (Prevention and Control of pollution) Act 1981 in short, was a law passed by the parliament of India to prevent and control the harmful effect of air pollution in India. This act is seen as the first concrete step taken by govt of India to combat air pollution.

Water pollution

water pollution may be defined as "any change in the physical, chemical and biological properties of water as well as contamination with any foreign substance, which would constitute a health hazard or otherwise decrease the utility of water".

It may also be defined as "addition of excess of undesirable substance to water".

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that make it harmful to man, animal and aquatic life or otherwise causes significant departures from the normal activities of various living communities in or around water.

Types of water pollution

1. ground water pollution
2. Surface water pollution
3. microbiological water pollution
4. Thermal water pollution
5. Oxygen depletion water pollution

Sources of water pollution

1. Domestic sewage:-

Major rivers of our country get polluted by the release of huge quantities of municipal and domestic wastes by drain. The sewage mainly contains human and animal waste, kitchen waste, street waste, garbage and organic substances like synthetic detergent which provide nutrition to bacteria, viruses and fungi.

Effect :- Disposal of sewage into rivers cause spreading of water borne diseases like dysentery, cholera, typhoid, hook - worm infections, malaria etc.

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2. Industrial waste:-

Water get polluted by acids, alkalis, detergents, soaps, Copper, zinc, lead, mercury, pesticides, insecticides and fungicides, which are released by chemical industries. pollution is also caused by wastes of industries like leather, sugar, paper, textile, steel, soap, oil refineries, pharmaceuticals etc.

Effect :-

Water containing such toxic substances cause poisoning of water, which damage the biological activity and kills useful organism.

3. Pollution due to oils:-

Oil and oil wastes enter rivers and other water bodies from different sources such as oil refineries, storage tanks, automobile waste, oil petro - chemical plant.

Effect :-

Oil being insoluble in water, floats and spreads rapidly into a thin layer on the surface of river or sea water which poses a serious threat to aquatic life. It also affects the flight and swimming capability of the sea birds.

4. Rain water:-

Natural rain water is slightly acidic but it becomes more acidic due to strong acids produced by industrial pollutants such as sulphuric acid and nitric acid which are produced by SO_2 or NO_2 respectively. Rain containing acid is called acid rain.

Effect:- Acid rain damages building material such as marble, lime stone etc. Aquatic animal and plant are also damaged.

5. Fertilizers and pesticides:-

Fertilizers, (nitrates and phosphates) pesticides, used in crop field are washed down by rain water to low lying areas, rivers or lakes causing water pollution.

Effect:- These salts help in the ^{growth} of other water pollutant like algae which de-oxygenate water disturbing aquatic life and produce foul smell.

6. Radio - active waste :-

Radio active substances can be carried

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into water from nuclear power plants and causes water pollution.

Effect:-

These substances may cause radioactivity in living organism and produce harmful effect.

7. Siltation:-

Suspended solids in water are sand, silt and minerals eroded from the land. Solid particles settle in reservoirs and dams and thus reduce their water storage capacity. They also block the sunlight required by bottom vegetation for photosynthesis.

8. Thermal Discharges:-

Water is used for cooling purposes in thermal and nuclear power plants. Cooling water is discharged at a raised temperature into the nearby flowing water. The increased temperature have several adverse effect on aquatic life.

9. Polychlorinated biphenyls

These compounds are used as fluids in transformers and capacitors. They are released in the atmosphere as vapours

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and mix with stain water and thus, contaminate water. They can cause cancer and skin disorders and hence are water pollution.

Characteristic of water pollution

Water pollution is characterised by the presence of excess physical, chemical and biological substances that change the qualities of the water and are capable of causing harm to living organism. We mentioned earlier that natural or unpolluted water is colourless, odourless and transparent.